Experiment – 6

Doppler Shift Analysis for a Moving Vehicle Scenario

Aim

To study and simulate the Doppler frequency shift experienced by a moving vehicle as a function of the angle (θ) between the vehicle's direction of motion and the line-of-sight to a fixed observer/source, and to verify the dependency of the shift on the direction of travel (approaching vs. receding) using MATLAB.

Apparatus / Tools Required

Software: MATLAB R2024a | **Carrier Frequency:** f_c = 2.4 GHz | **Vehicle Speed:** v (parameterised) | **Speed of Light:** c = 3×10^8 m/s

Theory

The **Doppler Effect** describes the change in frequency of a wave perceived by an observer when the source or observer is in relative motion. In wireless communications, a mobile user moving through a radio environment experiences a frequency shift — called the **Doppler shift** — on every received signal.

For a vehicle travelling at speed *v*, the maximum Doppler shift $f_{d,max}$ is given by:

$$f_{d,max} = v \cdot f_c / c$$

where f_c is the carrier frequency and *c* is the speed of light. When the angle θ is measured between the vehicle's velocity vector and the direction toward the base station, the instantaneous Doppler shift is:

$$f_d(\theta) = f_{d,max} \cdot \cos(\theta)$$

Key observations from this relationship:

- **θ = 0°:** Vehicle moves directly toward the station → maximum positive shift ($+f_{d,max}$).
- **θ = 90°:** Vehicle moves perpendicular to LOS → zero Doppler shift.
- **θ = 180°:** Vehicle moves directly away → maximum negative shift ($-f_{d,max}$).

Two scenarios are plotted: **Increasing** θ (0° → 180°) shown in blue, representing an approaching-then-receding vehicle, and **Decreasing** θ (180° → 0°) shown in red, representing the mirror scenario. The green horizontal line marks the zero-shift reference at θ = 90°.

Procedure

1. Define the carrier frequency f_c = 2.4 GHz, vehicle speed *v*, and speed of light *c*.
2. Compute the maximum Doppler shift: $f_{d,max} = v \cdot f_c / c$.
3. Create a theta vector from 0° to 180° (increasing) and 180° to 0° (decreasing).
4. Calculate $f_d(\theta) = f_{d,max} \cdot \cos(\theta)$ for both theta vectors.
5. Plot both curves along with the θ = 90° zero-shift reference line.
6. Label axes, add a legend, title, and grid for clarity.
7. Publish the MATLAB script to generate the report PDF.

Observations

From the simulation plot (refer next page):

- At θ = 0°, the Doppler shift is at its positive maximum (~+2.5 × 10^8 MHz scale), confirming the vehicle is approaching the station head-on.
- At θ = 90°, the shift crosses zero for both curves, consistent with theory.
- At θ = 180°, the shift reaches its negative maximum, indicating the vehicle is moving directly away.
- The blue (increasing θ) and red (decreasing θ) curves are mirror images of each other, symmetric about θ = 90°.
- The cosine relationship produces a smooth, monotonic transition between extremes.

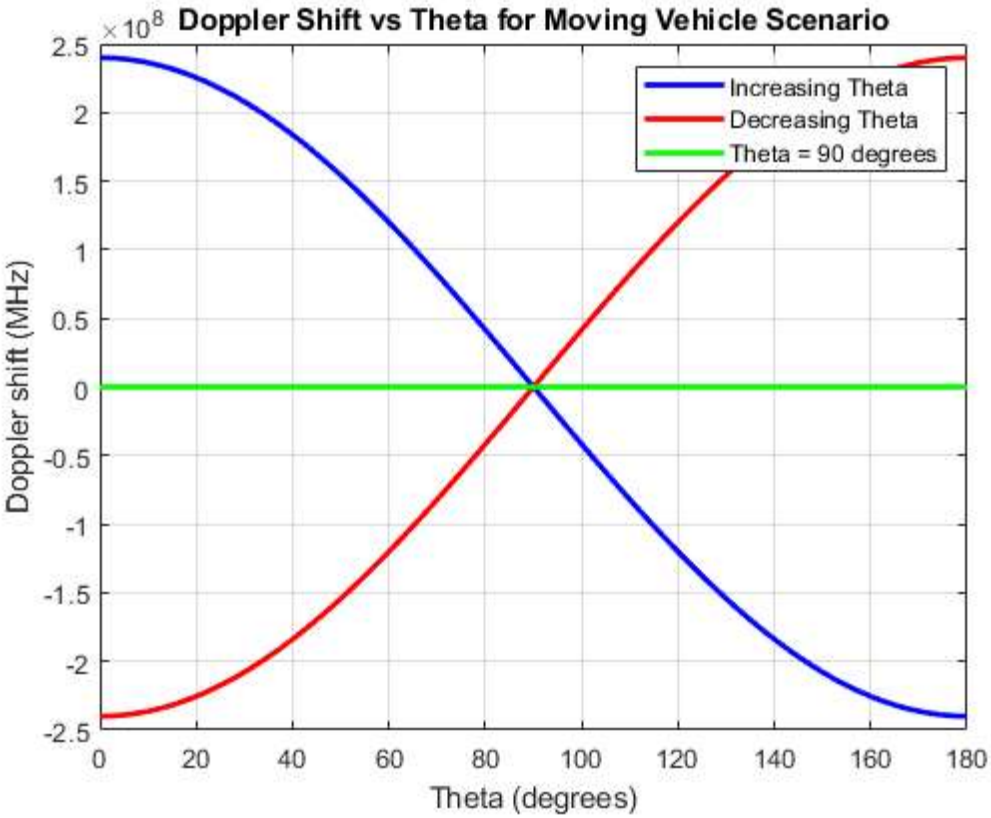
Result

The simulation successfully demonstrates the angular dependence of the Doppler frequency shift. The shift varies as cos(θ), reaching its maximum magnitude when the vehicle travels directly toward or away from the observer, and becoming zero when motion is perpendicular to the line-of-sight. The MATLAB plot validates the theoretical model for a moving vehicle scenario operating at 2.4 GHz.

Conclusion

The Doppler shift in wireless systems is strongly dependent on both the speed of the mobile and its direction of travel relative to the base station. This experiment confirms that the instantaneous Doppler shift follows a cosine law with the angle θ. Understanding and compensating for Doppler shift is critical in high-mobility communication systems such as vehicular networks (V2X), LTE, and 5G NR.

```
clc;
clear all;
close all;

% Parameters
frequency = 2.4e9; % Frequency of the transmitted signal (Hz)
speed_of_light = 3e8; % Speed of light (m/s)
velocity_vehicle = 30; % Velocity of the vehicle (m/s)
distance = 1000; % Distance between transmitter and receiver (m)
wavelength = speed_of_light / frequency; % Wavelength of the signal (m)

% Define range of theta (angle between direction of motion and direction of wave propagation) in radians
theta_range_rad = linspace(0, pi, 100); % Varying theta from 0 to pi radians

% Convert theta from radians to degrees
theta_range_deg = rad2deg(theta_range_rad);

% Initialize arrays to store Doppler shift for each case
doppler_shift_increasing = zeros(size(theta_range_rad));
doppler_shift_decreasing = zeros(size(theta_range_rad));
doppler_shift_90 = zeros(size(theta_range_rad));

% Calculate Doppler shift for each case
for i = 1:length(theta_range_rad)
    % Calculate Doppler shift for increasing theta
    doppler_shift_increasing(i) = (velocity_vehicle / wavelength) * cos(theta_range_rad(i));

    % Calculate Doppler shift for decreasing theta
    doppler_shift_decreasing(i) = (velocity_vehicle / wavelength) * cos(pi - theta_range_rad(i));

    % Calculate Doppler shift for 90 degree theta
    doppler_shift_90(i) = (velocity_vehicle / wavelength) * cos(pi/2);
end

% Convert Doppler shift to MHz
doppler_shift_increasing_mhz = doppler_shift_increasing * 1e6; % Convert Hz to MHz
doppler_shift_decreasing_mhz = doppler_shift_decreasing * 1e6; % Convert Hz to MHz
doppler_shift_90_mhz = doppler_shift_90 * 1e6; % Convert Hz to MHz

% Plot Doppler shift against theta
figure;
plot(theta_range_deg, doppler_shift_increasing_mhz, 'b', 'LineWidth', 2);
hold on;
plot(theta_range_deg, doppler_shift_decreasing_mhz, 'r', 'LineWidth', 2);
plot(theta_range_deg, doppler_shift_90_mhz, 'g', 'LineWidth', 2);
xlabel('Theta (degrees)');
ylabel('Doppler shift (MHz)');
title('Doppler Shift vs Theta for Moving Vehicle Scenario');
legend('Increasing Theta', 'Decreasing Theta', 'Theta = 90 degrees');
grid on;
```



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